



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CS8602 Compiler Design

Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic:** Unit III / Evaluation order for Syntax directed Definition

- **Course Outcome:** CO3

- **Topic Learning Outcome:** TLO8

- **Activity Chosen:** Reflection

- **Justification:**

The syntax directed definition concept is needed to understand how the type checking and intermediate code translation process will take place in compiler design. Hence reflection activity was planned on this topic to help the teacher to know how the students understood the particular concepts.

- **Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

- The activity was conducted in a classroom for a duration of 10 minutes.
- The topic of syntax directed definition & its evaluation order with examples were taught to the students.
- At the end of the class, the students were asked to write the questions in which they need more clarifications in the topic syntax directed definition.
- The students wrote and submit the questions they want to ask in a paper. It is shown in Fig.1.
- The teacher handed out the papers to the students in the following class after writing explanations for the questions on the papers.
- The student's questions are answered and cleared up after reading the responses.
- Many students had doubt on the same questions, and those questions were answered in class the next day for everyone.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO10	PO12	PSO1	PSO3
CO3	3	1	1	1	2	1	2	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO10	PSO1
	3	1	1	1	1

Justification for correlation	The students familiarize with the concept of syntax directed definition and apply this for translation of statement.	The students analyze the given grammar with rules to perform type checking and translation.	The students find the solutions for the given translation problems.	The students wrote their doubts clearly which improve their skills.	The students apply the SDD concepts in developing software compiler phases.
--------------------------------------	--	---	---	---	---

• **Images / Screenshot of the practice:**



Fig 1: Reflection activity

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- The sub topic that is unclear to many students can be easily identified
- The students were able to assess themselves and determine where they need more clarity.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- The student those who didn't understand the basics can be identified easily with this practice and they can be given more clarity in the next session which improve their level of understanding.

❖ **Challenges faced in implementation:**

- Some students didn't ask any questions, hence we cannot able to get the overall response from all the students.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/4_OIT551_Reflection.pdf
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/Reflection-HCI-MSS.pdf>

Signature of Faculty Member

HOD